GUI-G^2: Gaussian Reward Modeling For GUI Grounding

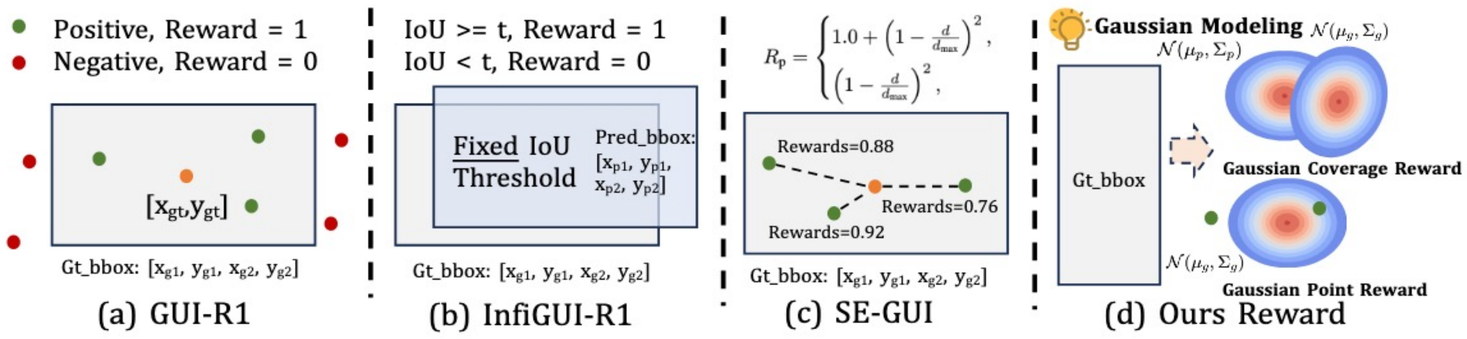
Fei Tang[1,2], Zhangxuan Gu[2], Zhengxi Lu[1], Xuyang Liu[2], Shuheng Shen[2†], Changhua Meng[2], Wen Wang[1], Wenqi Zhang[1] Yongliang Shen[1‡], Weiming Lu[1], Jun Xiao[1], Yueting Zhuang[1]

[1]Zhejiang University, [2]Ant Group ‡ : Corresponding author †: Project Leader

The 40th Annual AAAI Conference on Artificial Intelligence
JANUARY 20 – JANUARY 27, 2026 | SINGAPORE

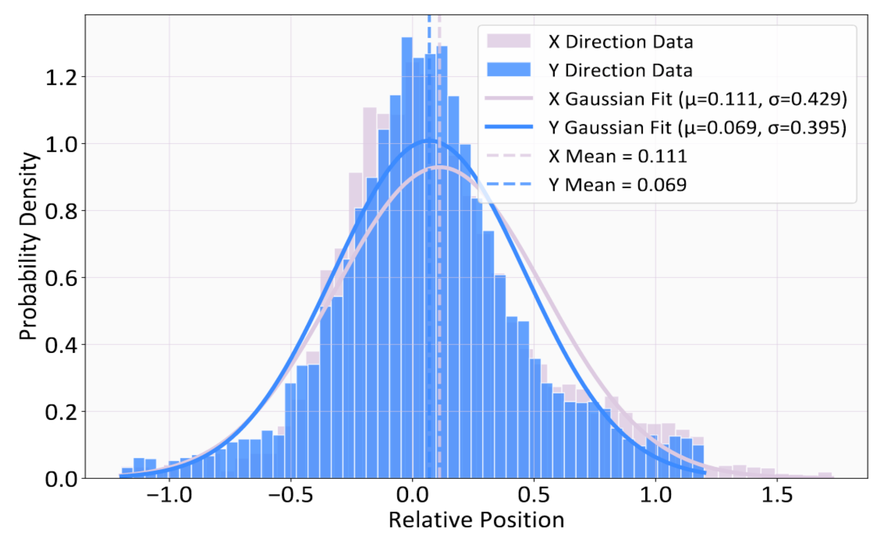
Introduction & Motivation

Previous Methods



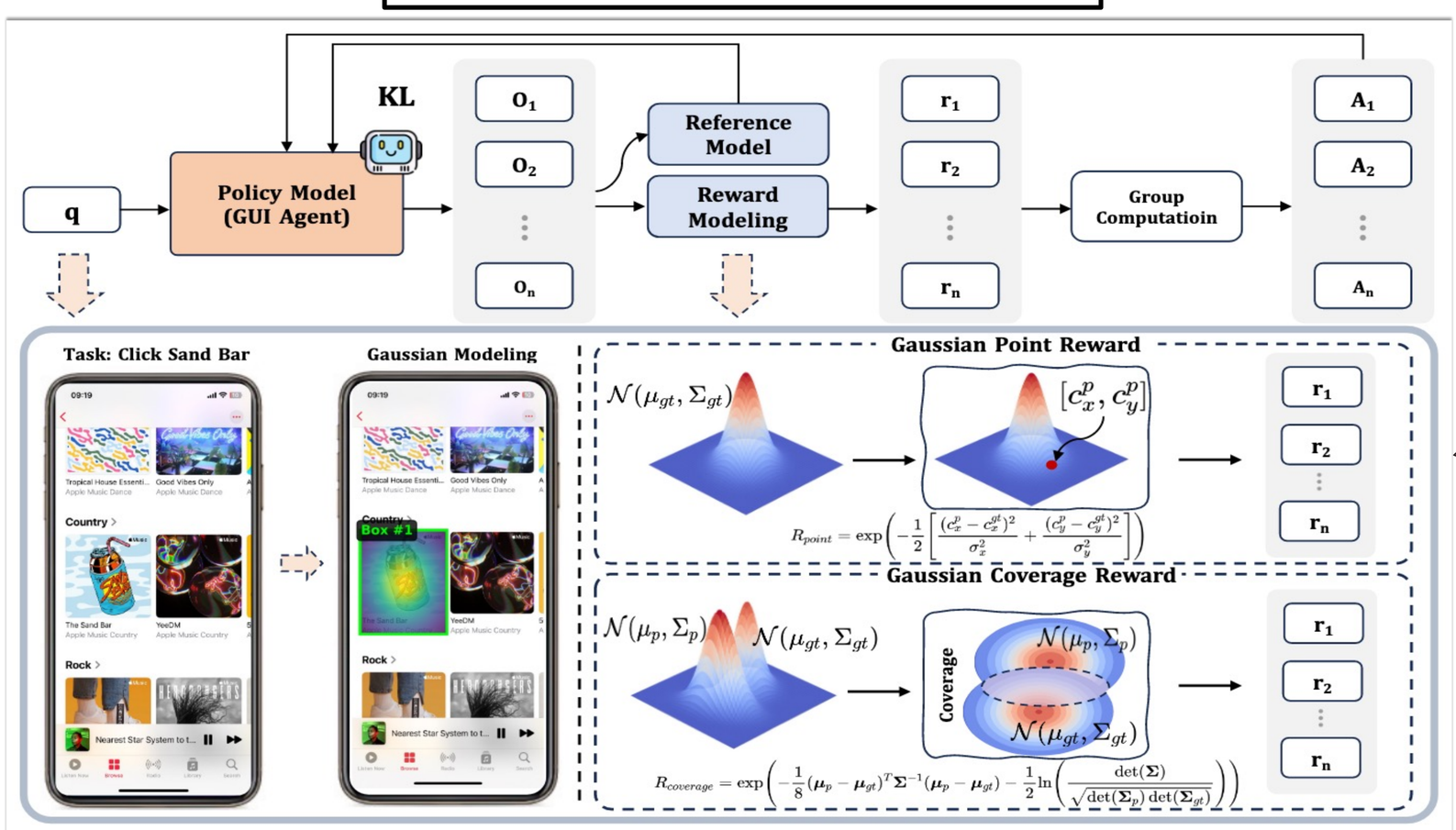
Existing RL methods rely on **binary "hit-or-miss" rewards**, providing zero guidance for "near-miss" predictions just pixels away from the target. Human clicking behavior naturally follows a **Gaussian distribution** around the center of target elements. (FROM AITW Dataset) →

Our Analysis

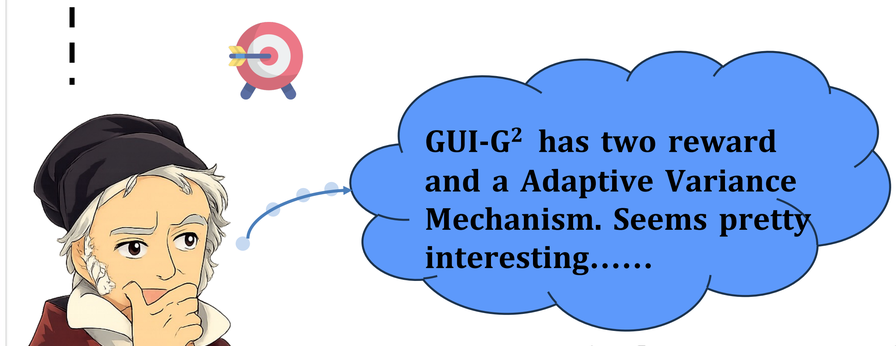


Our Method: GUI-G^2

GUI-G^2 Framework



Dual Gaussian Rewards: Combines **Point Rewards** and **Coverage Rewards** to model GUI elements as smooth distributions.



$$R_{point} = \mathcal{N}(\mu_p; \mu_{gt}, \Sigma_{gt}) = \exp\left(-\frac{1}{2}\left[\frac{(c_x^p - c_x^{gt})^2}{\sigma_x^{gt\,2}} + \frac{(c_y^p - c_y^{gt})^2}{\sigma_y^{gt\,2}}\right]\right)$$

$$R_{coverage} = \exp\left(-\frac{1}{8}(\mu_p - \mu_{gt})^T \Sigma^{-1} (\mu_p - \mu_{gt}) - \frac{1}{2}\ln\left(\frac{\det(\Sigma)}{\sqrt{\det(\Sigma_p)\det(\Sigma_{gt})}}\right)\right)$$

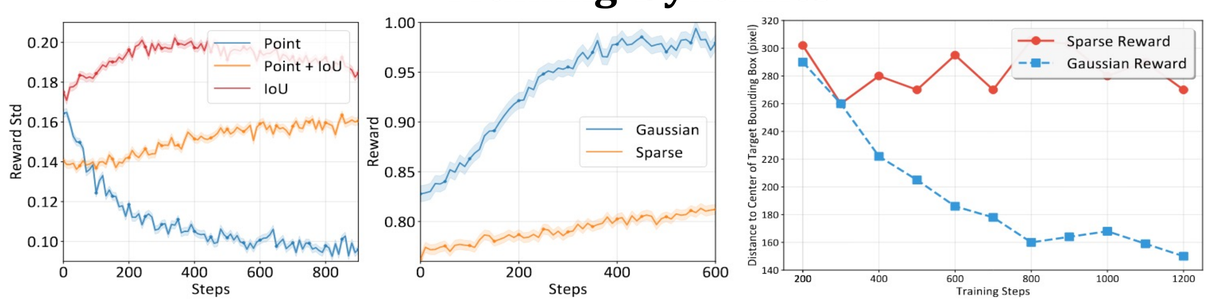
$$\sigma_x = \alpha \cdot (x_2 - x_1), \quad \sigma_y = \alpha \cdot (y_2 - y_1)$$

Adaptive Variance: Dynamically scales the Gaussian spread based on element size, ensuring consistent signals from **tiny icons to large panels**

Experiment

Model	CAD		Dev		Creative		Scientific		Office		OS		Avg.				
	Text	Icon	Text	Icon	Text	Icon	Text	Icon	Text	Icon	Text	Icon	Text	Icon	SS-Pro	SS-v1	SS-v2
Proprietary Models																	
GPT-4o	2.0	0.0	1.3	0.0	1.0	0.0	2.1	0.0	1.1	0.0	0.0	0.0	1.3	0.0	0.8	18.8	20.1
Claude Computer Use	14.5	3.7	22.0	3.9	25.9	3.4	33.9	15.8	30.1	16.3	11.0	4.5	23.4	7.1	17.1	83.0	-
General Open-source Models																	
Qwen2.5-VL-3B	9.1	7.3	22.1	1.4	26.8	2.1	38.2	7.3	33.9	15.1	10.3	1.1	23.6	3.8	16.1	55.5	80.9
Qwen2.5-VL-7B	16.8	1.6	46.8	4.1	35.9	7.7	49.3	7.3	52.5	20.8	37.4	6.7	38.9	7.1	26.8	84.7	88.8
GUI-specific Models (SFT)																	
SeeClick-9.6B	2.5	0.0	0.6	0.0	1.0	0.0	3.5	0.0	1.1	0.0	2.8	0.0	1.8	0.0	1.1	53.4	55.1
FOCUS-2B	7.6	3.1	22.8	1.7	23.7	1.7	25.0	7.1	23.2	7.7	17.8	2.5	19.8	3.9	13.3	-	-
CogAgent-18B	7.1	3.1	14.9	0.7	9.6	0.0	22.2	1.8	13.0	0.0	5.6	0.0	12.0	0.8	7.7	47.4	-
Aria-UI	7.6	1.6	16.2	0.0	23.7	2.1	27.1	6.4	20.3	1.9	4.7	0.0	17.1	2.0	11.3	-	-
OS-Atlas-7B	12.2	4.7	33.1	1.4	28.8	2.8	37.5	7.3	33.9	5.7	27.1	4.5	28.1	4.0	18.9	82.5	84.1
ShowUI-2B	2.5	0.0	16.9	1.4	9.1	0.0	13.2	7.3	15.3	7.5	10.3	2.2	10.8	2.6	7.7	75.1	77.3
UGround-V1-7B	15.8	1.2	51.9	2.8	47.5	9.7	57.6	14.5	60.5	13.2	38.3	7.9	45.2	8.1	31.1	-	-
UI-TARS-7B	20.8	9.4	58.4	12.4	50.0	9.1	63.9	31.8	63.3	20.8	30.8	16.9	47.8	16.2	35.7	89.5	91.6
UI-TARS-72B	18.8	12.5	62.9	17.2	57.1	15.4	64.6	20.9	63.3	26.4	42.1	15.7	50.9	17.6	38.1	88.4	90.3
JEDI-3B	27.4	9.4	61.0	13.8	53.5	8.4	54.2	18.2	64.4	32.1	38.3	9.0	49.8	13.7	36.1	-	88.6
JEDI-7B	38.0	14.1	42.9	11.0	50.0	11.9	72.9	25.5	75.1	47.2	33.6	16.9	52.6	18.2	39.5	-	91.7
GUI-Actor-7B	-	-	-	-	-	-	-	-	-	-	-	-	-	-	44.6	88.3	92.1
GUI-specific Models (RL)																	
UI-R1-3B	11.2	6.3	22.7	4.1	27.3	3.5	42.4	11.8	32.2	11.3	13.1	4.5	24.9	6.4	17.8	83.3	85.4
UI-R1-E-3B	37.1	12.5	46.1	6.9	41.9	4.2	56.9	21.8	65.0	26.4	32.7	10.1	-	-	33.5	89.2	89.5
GUI-R1-3B	26.4	7.8	33.8	4.8	40.9	5.6	61.8	17.3	53.6	17.0	28.1	5.6	-	-	-	-	-
GUI-R1-7B	23.9	6.3	49.4	4.8	38.9	8.4	55.6	11.8	58.7	26.4	42.1	16.9	-	-	-	-	-
InfiGUI-R1-3B	33.0	14.1	51.3	12.4	44.9	7.0	58.3	20.0	65.5	28.3	43.9	12.4	49.1	14.1	35.7	87.5	-
GUI-G1-3B	39.6	9.4	50.7	10.3	36.6	11.9	61.8	30.0	67.2	32.1	23.5	10.6	49.5	16.8	37.1	-	90.3
SE-GUI-3B	38.1	12.5	55.8	7.6	47.0	4.9	61.8	16.4	59.9	24.5	40.2	12.4	50.4	11.8	35.9	-	-
SE-GUI-7B	51.3	**42.2**	68.2	**19.3**	**57.6**	9.1	75.0	**28.2**	**78.5**	**43.4**	49.5	**25.8**	63.5	**21.0**	47.3	88.2	90.3
GUI-G^2-3B	38.1	**15.6**	**60.3**	**12.4**	**52.5**	9.1	**63.1**	20.0	63.0	19.2	41.1	6.7	**53.0**	13.1	**37.8**	**90.8**	**91.0**
GUI-G^2-7B	**55.8**	12.5	**68.8**	17.2	57.1	**15.4**	**77.1**	24.5	74.0	32.7	**57.9**	21.3	**64.7**	19.6	**47.5**	**92.0**	**93.3**

Training Dynamics



(a) Sparse reward training dynamics. (b) Distance convergence comparison.

Ablation Study

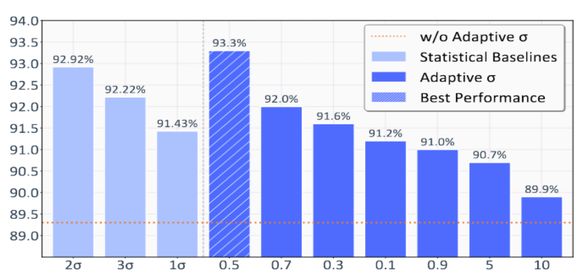


Figure 5: Hyperparameter sensitivity analysis for adaptive sigma (σ). Performance peaks at $\alpha = 0.5$ with 93.3% accuracy on Screenspot-v2.

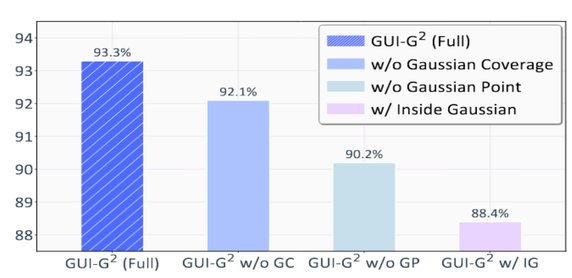


Figure 6: Ablation of Gaussian Component. Both Point and Coverage components contribute to the final 93.3% performance.

First author seeking internships, please feel free to reach out.

Add Wechat **Paper Link** **Model Link**